

Exercise 1. Evaluate

$$\int_1^{10} 5\sqrt{t} + \frac{2}{t^{1/3}} dt.$$

Exercise 2. Compute

$$\frac{d}{dx} \int_{\sin x}^{e^{x^2+1}} t^3 (\ln t) dt.$$

Exercise 3. Find the average value of $f(x) = \sqrt{9 - x^2}$ over $[-3, 3]$ and all $-3 \leq c \leq 3$ such that $f(c) = f_{\text{avg}}$.

Exercise 4. The following function represents the acceleration of a particle moving along the x -axis:

$$a(t) = 2t - 12 \text{ for } t \geq 0.$$

The starts at $x_0 = 3$ and has an initial velocity of $v_0 = 20$ units/sec.

- (a) Find the velocity function $v(t)$.
- (b) Find the position function $x(t)$.
- (c) Find the displacement on the time interval $[0, 4]$.
- (d) Find the total distance travel on the time interval $[0, 4]$.

Exercise 5. Evaluate

$$\int x \sin(\ln x) dx.$$

Exercise 6. Evaluate

$$\int (x^2 + x + 1) \ln x dx.$$

Exercise 7. Evaluate

$$\int \frac{1}{(e^t - e^{-t})} dt.$$

Exercise 8. Evaluate

$$\int (x^3 - x + 1)e^{2x} dx.$$

Exercise 9. Find the volume of the shape obtained by revolving the area between

$$f(x) = \sin x \text{ and } g(x) = \cos x$$

around the x -axis for $0 \leq x \leq \frac{\pi}{2}$.

Exercise 10. Find the area between

$$f(x) = \ln x \quad \text{and} \quad g(x) = x$$

for $1 \leq x \leq 10$.

Exercise 11. Evaluate

$$\int \frac{x}{x^2 - x - 2} dx.$$

Exercise 12. Evaluate

$$\int \frac{1}{x((\ln x)^2 - 3(\ln x) + 2)} dx.$$

Exercise 13. Determine whether the following improper integrals converge or diverge:

(a)

$$\int_1^{\infty} x^{100} e^{-x} dx$$

(b)

$$\int_1^{\infty} \frac{1}{x^{50} + 2x + 1} dx$$

(c)

$$\int_1^{\infty} \frac{\ln x}{x} dx$$

(d)

$$\int_1^3 \frac{1}{(x-2)^{1/4}} dx$$

Exercise 14. Solve for y with $y(0) = 1$:

(a)

$$\frac{dy}{dx} = (1 + x^3) \frac{y}{\ln y}$$

(b)

$$\frac{dy}{dx} = \frac{yx}{1 + x^2}$$

Exercise 15. Given a tank with 250L and 4kg of salt mixed in, suppose that 4L is added to the tank every minute which contains .25kg/L of salt. Moreover, 4L is drained out of the tank every minute so the amount of liquid remains constant. Find the concentration function for salt in the tank as a function of time.